FORECASTING U.S. MANUFACTURING STRESS

Duration: 7–10 Minutes **Format:** 2 Presenters **Focus:** AI Governance & Rare-Event Safety

2 Presenters	11 Slides	±0.03
~3.5 MIN EACH	TOTAL DECK COUNT	GUARDRAIL CAP

PART 1: PRESENTER 1 — PROBLEM, ARCHITECTURE & RESULTS (SLIDES 1–4)

SLIDE 1: TITLE & EXECUTIVE FRAMING (~30S)

> *"Good [morning/afternoon]. We are Team BMO Two, presenting Forecasting U.S. Manufacturing Stress with a Governed Adaptive Agent. Upfront, our honest framing: this was a bootcamp learning exercise to design, evaluate, and govern an agentic forecaster in finance—not to claim a production-ready trading model. Please keep that governance lens on everything we present today."*

SLIDE 2: BUSINESS PROBLEM & OPPORTUNITY (~40S)

> *"Credit risk teams need early warnings when an industry enters stress. But manufacturing stress is rare—occurring in only 20 out of 320 months between 2000 and 2026, clustered in 2008–09, 2020, and brief 2021–22 episodes. Our target definition: a month is in stress when U.S. manufacturing output—the FRED IPMAN index—falls 2% or more over 3 months, forecasted 3 months ahead. Combining rare historical events with opaque analyst judgment is the exact challenge a governed agent is built to address."*

SLIDE 3: HIGH-LEVEL APPROACH (~45S)

> *"Our solution pairs a transparent statistical model with a bounded LLM analyst: 1) A leakage-safe numerical anchor using logistic regression refit at each point in time. 2) Bounded LLM reasoning where the agent proposes a probability and rationale, but its adjustment to the anchor is strictly capped at ±0.03. 3) Governed adaptive memory where hypotheses require 3 distinct confirmations before altering strategy, with every change hash-linked."*

SLIDE 4: ARCHITECTURE & SYSTEM BOUNDARIES (~40S)

> *"Crucially, safety is enforced at the system boundary: The agent CAN adjust its proposed probability within the ±0.03 bound and refine its strategy text. The agent CANNOT alter underlying data, feature definitions, target definitions, or anchor code. No automated model retraining happens through the agent, ensuring zero data leakage and total code safety."*

PART 1 (CONT.): TECHNICAL SETUP & BACKTEST RESULTS (SLIDES 5–6)

SLIDE 5: DATA & BACKTEST SETUP (~30S)

> *"We evaluated our system using 15 monthly macroeconomic series from FRED and Yahoo Finance across 84 monthly backtest origins spanning 6 major stress episodes. Every origin strictly respects publication lags to prevent look-ahead bias."*

SLIDE 6: PRIMARY BACKTEST RESULTS (~45S)

> *"The headline result across all models? No model beat the historical-frequency baseline (Brier score 0.0667). Our bounded adaptive agent came in close at 0.0692, while the unbounded LLM analyst performed poorly at 0.0979. Why? Because macro signals 3 months out are largely coincident rather than leading. Handing over to Presenter 2 to explain how our guardrails saved performance."*

PART 2: PRESENTER 2 — GUARDRAILS & STRATEGY (SLIDES 7–9)

SLIDE 7: GUARDRAIL IN-DEPTH (~45S)

> *"Thanks Presenter 1. Did our safety guardrail work? Absolutely. Left unbounded, the LLM overreacted to transient news—proposing probabilities as high as 0.85 during volatile periods like mid-2020. However, our ±0.03 cap engaged on 7 critical origin dates, pulling the final forecast back near the baseline. The guardrail successfully limited harm, keeping our bounded agent tied near the anchor while unbounded variants failed."*

SLIDE 8: OUTCOME-GROUNDED ADAPTIVE MEMORY (~40S)

> *"We also tested outcome-grounded learning with strict temporal controls. Outcomes stay hidden until their official publication date, logging into a resolution ledger only when published. A dedicated reviewer evaluates resolved origins and updates a versioned strategy file (skill_state.yaml), recording timestamps, tools, and before/after hashes for complete auditability."*

SLIDE 9: NEXT STEPS & IMPROVEMENTS (~35S)

> *"To advance this proof of concept, our next steps are to: 1) Run a frozen prospective test with immutable pre-outcome logging. 2) Incorporate true leading indicators like new orders and survey indices. 3) Refine explanation quality and validate LLM rationales against feature snapshots."*

PART 2 (CONT.): VALUE, CONCLUSION & Q&A PREP (SLIDES 10–11)

SLIDE 10: POTENTIAL VALUE & INDUSTRY APPLICATIONS (~40S)

> *"The value of this work isn't just a forecasting metric—it's a reusable governance pattern for regulated finance: Combining a quantitative anchor + bounded LLM agent + immutable audit trail. It allows institutions to deploy generative AI safely with hard limits on behavior and full traceability."*

SLIDE 11: CONCLUSION & KEY TAKEAWAYS (~25S)

> *"In conclusion: 1) Always respect the base rate for rare events. 2) Anchor and bound your LLMs to prevent hallucination and overreaction. 3) System safety and governance can be proven even before predictive superiority is achieved. Thank you, and we welcome your questions!"*

Q&A FAST-REFERENCE CHEAT SHEET

Q1: "If no model beat the base rate, did the project fail?"

Answer: No. The project goal was to evaluate governance, safety, and agentic design. In rare-event macro forecasting (only 6.25% of months in stress), beating the historical frequency is notoriously difficult. We successfully proved that a bounded LLM prevents catastrophic overreaction while preserving a complete audit trail.

Q2: "Why was the guardrail cap set to ±0.03?"

Answer: It was established as a conservative parameter to keep the LLM anchored to the quantitative baseline. Backtesting confirmed its necessity: without the cap, the unbounded agent overreacted to noise and scored significantly worse (Brier score 0.0979 vs 0.0692).

Q3: "Did the agent actually learn over time?"

Answer: While predictive accuracy remained tied to the baseline, the agent proved out an auditable learning workflow. Across 24 reviews, 35 controlled mutations were safely executed in skill_state.yaml without breaking temporal controls or code boundaries.